Student t, df and mean CI when σ unknown

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Estimate population mean μ.
- Population σ unknown; sample s available.
- Population approximately normal under the course conditions.

2 DECISION / METHOD

1. Compute n, $\bar{X}$, s.
2. df=n-1.
3. t*=T.INV.2T(α,df).
4. SE=s/√n.
5. $\bar{X}$ ± t*SE; interpret subject to normality.

3 FORMULA

```
t* = T.INV.2T(α,df) > 0
CI: X¯ ± t* × s/√n
df=n-1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: s/√n and t_(n-1) when σ unknown. Still know: t critical lookup/function, normality condition and endpoint construction.

5 EXCEL ROUTE

- `n=COUNT(range)`; `Xbar=AVERAGE(range)`; `s=STDEV.S(range)`
- `tstar=T.INV.2T(alpha,n-1)`
- endpoints: `Xbar +/- tstar*s/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=25,s=4,n=16,C=.95. df=15,t*=2.131,SE=1 -> CI `(22.869,27.131)`.

7 TOP EXAM TRAPS

- df=n.
- Using Z because n seems moderate.
- Ignoring normality for a small sample.
- Mixing σ and s.

8 SAY ALOUD

Unknown σ means Student t with sample s and n-1 degrees of freedom.

9 EXIT CHECK

Which Excel function returns the positive two-sided t critical value?

ONE-GLANCE VISUAL ROUTE

